possible future observations with perfect certainty. The idea that “there’s something we don’t know,” given a certain wave function, is an outdated relic of our intuitive insistence that what we observe is what really exists. Quantum mechanics teaches us otherwise.

You’ll sometimes hear the idea, provoked by the uncertainty principle, that quantum mechanics *violates logic itself*. That’s silly. Logic deduces theorems from axioms, and the resulting theorems are simply true. The axioms may or may not apply to any given physical situation. Pythagoras’s theorem—the square of the hypotenuse of a right triangle equals the sum of the squares of the other two sides—is correct as a formal deduction from the axioms of Euclidean geometry, even though those axioms do not hold if we’re talking about curved surfaces rather than a flat tabletop.

The idea that quantum mechanics violates logic lives in the same neighborhood of the idea that atoms are mostly empty space (a bad neighborhood). Both notions stem from a deep conviction that, despite everything we’ve learned, particles are really points with some position and momentum, rather than being wave functions that are spread out.

Consider a particle in a box, where we’ve drawn a line dividing the box into left and right sides. It has some wave function that is spread throughout the box. Let proposition *P* be “the particle is on the left side of the box,” and proposition *Q* be “the particle is on the right side of the box.” We might be tempted to say that both of these propositions are false, since the wave function stretches over both sides of the box. But the proposition “*P* or *Q*” has to be true, since the particle is in the box. In classical logic, we can’t have both *P* and *Q* be false but “*P* or *Q*” be true. So something fishy is going on.

What’s fishy is neither logic nor quantum mechanics but our casual disregard for the nature of quantum states when assigning truth values to the statements *P* and *Q*. These statements are neither true nor false; they’re just ill defined. There is no such thing as “the side of the box the particle is on.” If the wave function were concentrated entirely on one side of the box and exactly vanished on the other, we could get away with assigning truth